A principle of balancing non-alternation and alternation in tonal metres

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Background

Regulated Verses 近体诗 (Wang 1962, Huang 2000):

- Flourished in Tang Dynasty (618 – 907 CE), during the Middle Chinese period.
- Four tones are divided into two categories:
  - Level (L): level (平)
  - Oblique (O): rising (上), falling (aka. 'departing' 去), checked (aka. 'entering' 入)
- Metrical templates **strictly** restrict what tone category can appear at every position in a line.
- Every line has 5 or 7 characters (= 5 or 7 syllables).
- Every poem has 4 or 8 lines. Every 4 lines are divided into 2 couplets.
- The second half of an 8-line poem repeat the template of the first 4 lines.

Metrics

(1) Template			initial tone L		initial tone O
length	5	couplet 5L	L L L O O O O O L L	couplet 5O	O O L L O L L O O L
		couplet 5O	O O L L O L L O O L	couplet 5L	L L L O O O O O L L
	7	couplet 7L	L L O O L L O O O L L O O L	couplet 7O	O O L L L O O L L O O O L L
		couplet 7O	O O L L L O O L L O O O L L	couplet 7L	L L O O L L O O O L L O O L

- Templates can be divided into 4 types depending on…
  - the tone of the first character in the first line: L-initial and O-initial
  - the number of characters in a line: 5 or 7
- There are 4 types of couplets (2 line lengths × 2 initial tones): 5L, 5O, 7L, 7O.
- Final characters (grey) must use the tone category as specified. Since it's already the most strongly regulated position, restrictions to other positions no longer apply.
- Only blue positions allow inversions.
  → Inversion: using the opposite tone category required by the template.
- 7-syllables lines are 'extensions' of 5-syllable lines, because
  - couplet 7O = OO + couplet 5L
  - couplet 7L = + LL + couplet 5O

So, I will focus on the last 5 characters of lines.

Observations

Alternation and non-alternation must both exist!

Alternation 1: Regulated Verses more strongly regulate even-numbered positions (i.e., no inversions in these positions) to …

a. yield an iambic metre (Kiparsky, 2020); and

b. to avoid monotony (since the tone categories in these positions are usually different in each line).

Alternation 2: The number of characters from the same tone category in a row never exceeds 3.

'Non-alternation': Non-final characters in each line always have at least one neighbour from the same tone category.

Case Study

'No Lonely Level Tone' 孤平

A line cannot have only one character from the L tone category.

- There are two positions (yellow) that could have allowed inversions because they are odd-numbered, but do not in reality. They are:
  - the first character of the second line in couplet 5O, and
  - the third character of the second line in couplet 7L.
- In traditional Chinese literature, these exceptions are said to avoid a phenomenon called *Lonely L*.
- Because, if these positions allowed inversions, the entire line might end up with only one L-tone character (red below) besides the final position.
- If inversions at these positions cannot be avoided, a possible remedy is to 'change the tone category of the character immediately following this L' (green below).

(2)	with lonely L	solution
5O	O L O O L	O L L O L
7L	O O O L O O L	O O O L L O L

Finding

This can be reduced to simultaneous alternation and non-alternation.

(3) Constraint	Assign one violation for …
BIN-L	each non-final L without an L neighbour.
BIN-O	each non-final O without an O neighbour.
ODD-INVER	each inversion occurring in an even-numbered position.
NO-INVER-FIN	each inversion occurring in the line-final position.

If we take …

- the optimal template as the UR, and
- the possible lines SFs after changes as candidates,

the solution to *Lonely L* can be derived using the constraints above.

- BIN-L and BIN-O are the key constraints for non-alternation
  → the *Lonely-L* case is penalised for lacking binarity.
- ODD-INVER ensures that, even if an L ends up having no neighbour from the same category, remedies should be sought in odd-numbered positions only
  → removing the non-binary L (4c) or adding an L (4d) is not a solution.
- NO-INVER-FIN means that
  → no inversion should be allowed at line-final positions,
  → 'LOL' at the end of a line should stay as it is, even though non-alternation is violated,
  → so, don't change to 'LOO'.
- Seven-syllable lines work analogously (5).

Implications

1. **Shows simultaneous requirement of alternation and non-alternation …**
   - in the surface pattern of the optimal template itself; and
   - *where* and *to what extent* the templates allow variations / inversions.
2. **Enriches the typology of metres and metrical principles**
   - stress / accentual / quantitative metres: prefer alternating prominence
     → WSWS where a position is surrounded by two items from the opposite categories;
   - Middle Chinese Regulated Verses: prefer gradual change of tonal categories
     → BIN-L and BIN-O: ideally *not* surrounded by two items from the other category.
3. **Demonstrates the mutual influence of metres and phonological theories of languages**

non-alternation	⇔	even when looking at abstracted higher-level categories, the tendency for tones to spread across multiple TBUs (Hyman 2011)
at least two adjacent syllables from the same tone category	⇔	the generalisation that many tonal languages avoid rapid excursions of pitch in a short time (McPherson 2016; cf. *HLH, Cahill 2007; *TROUGH, Yip 2002)
alternation	⇔	The 'up-to-three' pattern on the number of syllables from the same tone category resembles the limit that tonal spreading usually does not extend beyond two adjacent TBUs (e.g., Prinmi, Jardin 2019; Dembwa Taita, Odden 2001)

Conclusions

The optimal template shape of Regulated Verses is not merely a choice to the poets' liking, but the most balanced option under the tension between natural-language tendency of tones to spread vs. the rhythmic alternation of poetry.

(4)	LLOOL	BIN-L	ODD-INVER	NO-INVER-FIN	BIN-O
a. ☞	OLLOL				**
b.	OLOOL	*!			*
c.	OOOOL		*!		
d.	OLOLL	*!	*		**
e.	OLLOO			*!	*
(5)	**OOLLOOL**	**BIN-L**	**ODD-INVER**	**NO-INVER-FIN**	**BIN-O**
a. ☞	OOOLLOL				*
b.	OOOLOOL	*!			
c.	OOOOOOL		*!		
d.	OOOLOLL	*!	*		*
e.	OOOLLOO			*!	

Notes:

There is another method of repairing *Lonely L*, which requires the character at the same position of the other line in the same couplet to be inverted. More constraints are needed to derive such a pattern and sublinearity might be involved. Research is still ongoing and let me know if you are interested!

Selected References: Cahill, Mike. 2007. More Universals of Tone. *SIL Electronic Working Papers* 72. · Huang, Ji. 2000. *Shi Ci Xue Bu 诗词学步 (Introduction to the Metrics of Poems and Ci)*. 1st ed. Beijing: Beijing shi fan da xue chu ban she 北京师范大学出版社. · Hyman, Larry M. 2011. Tone: Is it different? In *The Handbook of Phonological Theory*, ed. John Goldsmith, Jason Riggle, and Alan C. L. Yu, 2nd ed., 197–239. Oxford: Blackwell Publishing Ltd. · Jardine, Adam. 2020. Melody learning and long-distance phonotactics in tone. *Natural Language & Linguistic Theory* 38: 1145–1195. · Kiparsky, Paul. 2020. Metered Verse. *Annual Review of Linguistics* 6: 25–44. · McPherson, Laura. 2016. Cumulativity and Ganging in the Tonology of Awa Suffixes. *Language: Phonological Analysis* 92(1): e38-e66. · Odden, David. 2001. Tone shift and spread in Taita I. *Studies in African Linguistics* 30: 75–110. · Wang, Li. 1962. *Shi Ci Ge Lü 诗词格律 (Poetic Metrics)*. Beijing: Zhong Hua Shu Ju 中华书局. · Yip, Moira. 2002. *Tone*. Cambridge: Cambridge University Press.